

[20 marks]

Exercise I.

Compute the following integrals:

1. $\int \frac{\sin^2 x}{\cos^4 x} dx,$

2. $\int \frac{x^4}{1+x^2} dx,$

3. $\int x \arctan \sqrt{x} dx,$

4. $\int \frac{x}{\sqrt{(1-x)(x)}} dx.$

Exercise II.

Let $\alpha \in \mathbb{R}$. Determine whether the following series converges absolutely, converges or diverges:

[20 marks]

1. $\sum_{n=2}^{\infty} \frac{(-1)^n}{n \ln n},$

2. $\sum_{n=2}^{\infty} \frac{\alpha^n}{n \ln n},$

3. $\sum_{n=2}^{\infty} \frac{(-1)^n + 1}{n},$

4. $\sum_{n=4}^{\infty} \frac{(-1)^n}{n + (-1)^n}.$

Hint: The sequence of general term $b_n = \frac{1}{n + (-1)^n}$ is not decreasing.

Exercise III.

[10 marks]

Find $\lim_{n \rightarrow +\infty} \sum_{k=1}^n \frac{k}{n^2} \ln(1 + \frac{k^2}{n^2}).$

Exercise IV.

[20 marks]

Let $\alpha \in \mathbb{R}$. Study the natures of the following improper integrals:

1. $\int_1^{\infty} \frac{1}{(\arctan x)^2} dx,$

2. $\int_0^1 \frac{\sin x}{\sqrt{|\ln x|}} dx,$

3. $\int_0^{\infty} |\alpha|^x dx.$

Exercise V.

[15 marks]

Let $\alpha > 0$ and $n \in \mathbb{N}$.

1. Show that $\lim_{n \rightarrow \infty} \int_{2\pi n}^{2\pi n + \frac{\pi}{4}} x^\alpha \cos x dx = \infty.$

2. Deduce the nature of $\int_1^{\infty} x^\alpha \cos x dx.$

Exercise VI.

[15 marks]

Let $I_n = 2 \int_0^{\frac{\pi}{2}} \cos^{2n} t dt$ for all $n \in \mathbb{N}$.

1. Show that $2nI_n = (2n-1)I_{n-1}$ for $n \geq 1$.

2. Determine the value of I_n in terms of n .

3. Show that $\int_0^1 \frac{x^{n-\frac{1}{2}}}{\sqrt{1-x}} dx = I_n.$

Ex I. compute the following integrals:

$$1. \int \frac{\sin^2 x}{\cos^4 x} dx = \int \tan^2 x \cdot \frac{1}{\cos^2 x} dx; \quad \tan x = t \Rightarrow \frac{1}{\cos^2 x} dx = dt.$$

$$= \int t^2 dt = \frac{t^3}{3} + c = \frac{\tan^3 x}{3} + c.$$

$$2. \int \frac{x^4}{1+x^2} dx = \int \left[x^2 - 1 + \frac{1}{x^2+1} \right] dx = \frac{x^3}{3} - x + \arctan x + c.$$

$$3. \int x \arctan \sqrt{x} dx = 2 \int t^3 \arctan t dt; \quad \sqrt{x} = t \Rightarrow dx = 2t dt$$

$$= 2 \left[\frac{t^4}{4} \arctan t - \frac{1}{4} \int \frac{t^4}{1+t^2} dt \right]; \quad u' = t^3 \Rightarrow u = \frac{t^4}{4}$$

$$v = \arctan t \Rightarrow v' = \frac{1}{1+t^2}$$

$$= \frac{t^4}{2} \arctan t - \frac{1}{6} t^3 + \frac{1}{2} t - \frac{1}{2} \arctan t + c$$

$$\stackrel{t=\sqrt{x}}{=} \frac{x^2}{2} \arctan \sqrt{x} - \frac{1}{6} x^{3/2} + \frac{1}{2} x^{1/2} - \frac{1}{2} \arctan \sqrt{x} + c. \quad \text{(Bonus)}$$

Or one can proceed as follows: integ by parts, then substitution.

$$\int \underbrace{x}_{u'} \underbrace{\arctan \sqrt{x}}_v dx = \frac{x^2}{2} \arctan \sqrt{x} - \frac{1}{4} \int x^2 \cdot \frac{1}{\sqrt{x}(1+x)} dx; \quad u' = x \Rightarrow u = \frac{x^2}{2}$$

$$v = \arctan \sqrt{x} \Rightarrow v' = \frac{1}{2\sqrt{x}(1+x)}$$

$$= \frac{x^2}{2} \arctan \sqrt{x} - \frac{1}{2} \int \frac{t^4}{1+t^2} dt \Big|_{t=\sqrt{x}}; \quad \sqrt{x} = t \Rightarrow \frac{1}{2\sqrt{x}} dx = dt$$

$$= \frac{x^2}{2} \arctan \sqrt{x} - \frac{1}{6} t^3 + \frac{1}{2} t - \frac{1}{2} \arctan t \Big|_{t=\sqrt{x}} + c$$

$$= \frac{x^2}{2} \arctan \sqrt{x} - \frac{1}{6} x^{3/2} + \frac{1}{2} x^{1/2} - \frac{1}{2} \arctan \sqrt{x} + c. \quad \text{(Bonus)}$$

$$4. \int \frac{x}{\sqrt{(1-x)x}} dx = \int \frac{x}{\sqrt{x^2 - x}} dx = \int \frac{x}{x\sqrt{1-\frac{1}{x}}} dx; \quad \frac{1-x}{x} = t^2 \Rightarrow x = \frac{1}{1+t^2} \quad \& \quad dx = \frac{-2t}{(1+t^2)^2} dt.$$

$$\left[t = \sqrt{\frac{1-x}{x}} \right]$$

$$= \int \frac{1}{t} \cdot \frac{-2t}{(1+t^2)^2} dt = -2 \int \frac{dt}{(1+t^2)^2} = -2 I_2; \quad \text{I}_2 \text{ may done by 2 ways.}$$

$$\star \quad I_2 = \int \frac{dt}{(1+t^2)^2} = \int \frac{(1+\tan^2 \theta)}{(1+\tan^2 \theta)^2} d\theta; \quad t = \tan \theta \Rightarrow dt = (1+\tan^2 \theta) d\theta$$

$$= \int \frac{1}{1+\tan^2 \theta} d\theta = \int \cos^2 \theta d\theta = \int \frac{1+\cos 2\theta}{2} d\theta = \frac{1}{2} \theta + \frac{1}{4} \sin(2\theta) \Big|_{\theta=\arctan t} + c$$

☆ we integrate $I_1 = \int \frac{dt}{(1+t^2)^2} dt = \arctan t$ by parts:

$$I_1 = \int \underbrace{1}_{u'} \cdot \underbrace{\frac{1}{(1+t^2)^2}}_v dt = t \cdot \frac{1}{1+t^2} - \int t \cdot \frac{-2t}{(1+t^2)^2} dt = \frac{t}{1+t^2} + 2 \int \frac{t^2-1}{(1+t^2)^2} dt$$

$$= \frac{t}{1+t^2} + 2 I_1 - 2 I_2.$$

$$\text{Hence } I_2 = \frac{1}{2} \left[\frac{t}{1+t^2} + I_1 \right] = \frac{1}{2} \left[\frac{t}{1+t^2} + \arctan t \right].$$

2nd way
for I_2

Another way to calculate $\int \frac{x}{\sqrt{(1-x)x}} dx$:

$$\int \frac{x}{\sqrt{(1-x)x}} dx = \int \frac{x}{\sqrt{-[x^2-x]}} dx \stackrel{(1)}{=} \int \frac{x}{\sqrt{-[x^2-x+\frac{1}{4}-\frac{1}{4}]}} dx$$

$$\stackrel{(1)}{=} \int \frac{x}{\sqrt{-(x-\frac{1}{2})^2 + (\frac{1}{2})^2}} dx \quad ; \quad x - \frac{1}{2} = \frac{1}{2} \sinh \theta \stackrel{(1)}{\Rightarrow} dx = \frac{1}{2} \cosh \theta d\theta$$

$$\stackrel{(1)}{=} \int \frac{\frac{1}{2} + \frac{1}{2} \sinh \theta}{\sqrt{-\frac{1}{4} \sinh^2 \theta + \frac{1}{4}}} \left(\frac{1}{2} \cosh \theta \right) d\theta$$

$$\stackrel{(1)}{=} \left[\frac{1}{2} \theta - \frac{1}{2} \cosh \theta \right]_{\arcsinh(2x-1)} + C$$

Another possible substitution

$$* x - \frac{1}{2} = \frac{1}{2} \cosh \theta$$

$$* x - \frac{1}{2} = \frac{1}{2} \tanh \theta$$

Ex II. Let $x \in \mathbb{R}$. Determine whether the following series converges absolutely, converges or diverges:

1. $\sum_{n=2}^{\infty} \frac{(-1)^n}{n \ln n}$

☆ $\sum_{n=2}^{\infty} \left| \frac{(-1)^n}{n \ln n} \right| \stackrel{(1)}{=} \sum_{n=2}^{\infty} \frac{1}{n \ln n}$ div (Bertrand $\alpha=1 \wedge \beta=1 \leq 1$). (1)

☆ $\sum_{n=2}^{\infty} \frac{(-1)^n}{n \ln n}$ conv, Alt-Series test $\left\{ \frac{1}{n \ln n} \right\} \stackrel{(1)}{\text{dec}} \& \stackrel{(1)}{\rightarrow 0}$.

2. $\sum_{n=2}^{\infty} \frac{x^n}{n \ln n}$

☆ Ratio test:

$$\left| \frac{a_{n+1}}{a_n} \right| = \left| \frac{x^{n+1}}{(n+1) \ln(n+1)} \cdot \frac{n \ln n}{x^n} \right| \stackrel{(1)}{=} |x| \frac{n \ln n}{(n+1) \ln(n+1)} \xrightarrow{n \rightarrow \infty} |x| \stackrel{(1)}{}$$

• If $|x| < 1$, the series converges absolutely. (1)

• If $|x| > 1$, the series diverges. (1)

• The case $|x| = 1$ (i.e. $x = \pm 1$) done part 1. (1)

☆ Cauchy's Root test:

$$|a_n|^{1/n} \stackrel{(1)}{=} \frac{|\alpha|}{(n \ln n)^{1/n}} = \frac{|\alpha|}{e^{\frac{1}{n} \ln(n \ln n)}} = \frac{|\alpha|}{e^{\frac{1}{n} [\ln n + \ln \ln n]}} \xrightarrow{n \rightarrow \infty} |\alpha| \stackrel{(1)}{>} |\alpha|$$

• If $|\alpha| < 1$, the series converges absolutely. (1)

• If $|\alpha| > 1$, the series diverges. (1)

• The case $|\alpha| = 1$ (i.e. $\alpha = \pm 1$) done part 1. (1)

$$3. \sum_{n=2}^{\infty} \frac{(-1)^n + 1}{n}$$

$$\left. \begin{array}{l} \sum_{n=2}^{\infty} \frac{(-1)^n}{n} \text{ conv (Alt. ser. test)} \\ \sum_{n=2}^{\infty} \frac{1}{n} \text{ div } (\text{Riemann } \alpha = 1 \leq 1) \end{array} \right\} \stackrel{(1)}{=} \sum_{n=2}^{\infty} \frac{(-1)^n + 1}{n} \text{ div.}$$

$$4. \sum_{n=4}^{\infty} \frac{(-1)^n}{n + (-1)^n} \text{ Alt; } b_n = \frac{1}{n + (-1)^n} \text{ is not dec.}$$

$$\star \sum_{n=4}^{\infty} \left| \frac{(-1)^n}{n + (-1)^n} \right| = \sum_{n=4}^{\infty} \frac{1}{n + (-1)^n} \text{ Divergent (1)}$$

• Equival test $\frac{1}{n + (-1)^n} \sim \frac{1}{n} > 0$ & $\sum \frac{1}{n}$ div.

or • Comparison test $\frac{1}{n + (-1)^n} \geq \frac{1}{n + n} \geq \frac{1}{2n} > 0$ & $\frac{1}{2} \sum \frac{1}{n}$ div.

$$\star \sum_{n=4}^{\infty} \frac{(-1)^n}{n + (-1)^n}$$

$$\frac{(-1)^n}{n + (-1)^n} = \frac{(-1)^n}{n} \left[\frac{1}{1 + \frac{(-1)^n}{n}} \right] = \frac{(-1)^n}{n} \left[1 - \frac{(-1)^n}{n} + o\left(\frac{(-1)^n}{n}\right) \right]$$

$$= \frac{(-1)^n}{n} - \frac{1}{n^2} + o\left(\frac{1}{n^2}\right); \quad (1)$$

$$\sum \frac{1}{n^2} \text{ conv absd (Riemann)} \stackrel{(1) \text{ Bonus}}{=} \sum o\left(\frac{1}{n^2}\right) \text{ conv absd.}$$

$$\text{Now } \sum \frac{(-1)^n}{n} \text{ conv} \wedge \sum \frac{1}{n^2} \text{ conv} \wedge \sum o\left(\frac{1}{n^2}\right) \text{ conv} \stackrel{(1)}{=} \sum \frac{(-1)^n}{n + (-1)^n} \text{ conv}$$

Ex III. Find $\lim_{n \rightarrow \infty} \sum_{k=1}^n \frac{k}{n^2} \ln(1 + \frac{k^2}{n^2})$

$$\sum_{k=1}^n \frac{k}{n^2} \ln(1 + \frac{k^2}{n^2}) = \frac{1}{n} \sum_{k=1}^n \frac{k}{n} \ln(1 + (\frac{k}{n})^2)$$

$$= \frac{1}{n} \sum_{k=1}^n f(k/n); f(x) = x \ln(1+x^2)$$

$$\xrightarrow{n \rightarrow \infty} \int_0^1 f(x) dx \quad \left| \begin{array}{l} \bullet 1+x^2 > 0 \\ \bullet f \in C([0,1]) \subseteq R([0,1]) \end{array} \right.$$

$$= \int_0^1 x \ln(1+x^2) dx \quad \left| \begin{array}{l} \bullet u = x^2 \text{ \& } v = \ln(1+x^2) \\ \bullet \text{ are } C^1 \text{ fns on } [0,1] \end{array} \right.$$

$$= \frac{x^2}{2} \ln(1+x^2) \Big|_0^1 - \int_0^1 \frac{x^3}{1+x^2} dx \quad \left| \begin{array}{l} \bullet v' = \frac{2x}{1+x^2} \end{array} \right.$$

$$= \frac{1}{2} \ln 2 - \int_0^1 \left[x - \frac{x}{x^2+1} \right] dx \quad \left| \begin{array}{l} \frac{x^3}{x^2+x} \Big| \frac{x^2+1}{x} \end{array} \right.$$

$$= \frac{1}{2} \ln 2 - \int_0^1 x dx + \int_0^1 \frac{2x}{x^2+1} dx$$

$$= \frac{1}{2} \ln 2 - \left[\frac{x^2}{2} \right]_0^1 + \left[\frac{1}{2} \ln(x^2+1) \right]_0^1$$

$$= -1/2 + \ln 2.$$

Ex IV. Let $\alpha \in \mathbb{R}$. Study the natures of the following imp. integrals

1. $\int_1^{\infty} \frac{1}{(\arctan x)^2} dx$ Div,

☆ equivalent test $\frac{1}{(\arctan x)^2} \sim \frac{1}{(\pi/2)^2} > 0$ & $\int_1^{\infty} \frac{1}{(\pi/2)^2} dx$ div.

☆ comparison test $\frac{1}{(\arctan x)^2} \geq \frac{1}{(\pi/2)^2} > 0$ & $\int_1^{\infty} \frac{1}{(\pi/2)^2} dx$ div

☆ x^α test $x^{1/2} \cdot \frac{1}{(\arctan x)^2} \xrightarrow{x \rightarrow \infty} \infty$ & $\alpha = 1/2 \leq 1$.

$$2. \int_{\boxed{0}}^{\boxed{1}} \frac{\sin x}{\sqrt{|\ln x|}} dx \stackrel{\text{formally}}{=} \int_{\boxed{0}}^{\boxed{1/2}} \frac{\sin x}{\sqrt{|\ln x|}} dx + \int_{\boxed{1/2}}^{\boxed{1}} \frac{\sin x}{\sqrt{|\ln x|}} dx = I_1 + I_2$$

near 0^+ :

$$\star \frac{\sin x}{\sqrt{|\ln x|}} \stackrel{\textcircled{1}}{\sim}_{0^+} \frac{x}{|\ln x|^{1/2}} = \frac{1}{x^{-1} |\ln x|^{1/2}} \stackrel{\textcircled{1}}{>} 0$$

$$\int_0^{1/2} \frac{1}{x^{-1} |\ln x|^{1/2}} dx \text{ conv (Bat-d)} \stackrel{\textcircled{1}}{\Rightarrow} I_1 \text{ conv (equiv test)}$$

$$\text{or } \star 0 \stackrel{\textcircled{1}}{<} \frac{\sin x}{|\ln x|^{1/2}} \stackrel{\textcircled{1}}{<} \frac{x}{|\ln x|^{1/2}}$$

$$\int_0^{1/2} \frac{1}{x^{-1} |\ln x|^{1/2}} dx \text{ conv (Bat-d)} \stackrel{\textcircled{1}}{\Rightarrow} I_1 \text{ conv (compted)}$$

$$\text{or } \star 0 \stackrel{\textcircled{1}}{<} \frac{\sin x}{|\ln x|^{1/2}} \stackrel{\textcircled{1}}{<} \frac{1}{|\ln x|^{1/2}}$$

$$\int_0^{1/2} \frac{1}{x^0 |\ln x|^{1/2}} dx \text{ conv (Bat-d)} \stackrel{\textcircled{1}}{\Rightarrow} I_1 \text{ conv (compted)}$$

near 1^- :

$$\star \frac{\sin x}{\sqrt{|\ln x|}} \stackrel{\textcircled{2}}{\sim}_{1^-} \frac{\sin 1}{|x-1|^{1/2}} > 0$$

$$\int_{1/2}^1 \frac{1}{|x-1|^{1/2}} dx \text{ conv (Riemann)} \stackrel{\textcircled{1}}{\Rightarrow} I_2 \text{ conv (equiv test)}$$

$$\text{or } \star (1-x)^{3/4} \cdot \frac{\sin x}{\sqrt{|\ln x|}} \stackrel{\textcircled{3}}{\xrightarrow{x \rightarrow 1}} > 0 \text{ \& } \alpha = \frac{3}{4} < 1 \stackrel{\textcircled{1}}{\Rightarrow} I_2 \text{ conv (x test)}$$

$$\text{Conclusion: } I_1 \text{ conv} \wedge I_2 \text{ conv} \Rightarrow \int_0^1 \frac{\sin x}{\sqrt{|\ln x|}} dx \text{ conv.}$$

$$3. \int_0^{\infty} |x|^{\alpha} dx = I_{\alpha}$$

$$\begin{cases} \star \alpha = 1: \textcircled{1} I_1 = \int_0^{\infty} 1 dx \text{ div} \\ \star \alpha > 1: \textcircled{3} x^0 \cdot |x|^{\alpha} = e^{x \ln |x|} \xrightarrow{x \rightarrow \infty} \infty, \quad 0 \leq 1 \Rightarrow I_{\alpha} \text{ div} \\ \star \alpha < 1: \textcircled{3} x^{10} \cdot |x|^{\alpha} = x^{10} e^{x \ln |x|} \xrightarrow{x \rightarrow \infty} 0 \quad |0| > 1 \Rightarrow I_{\alpha} \text{ conv} \end{cases}$$

$$\star \bullet \alpha = 1: \textcircled{1} I_1 = \int_0^{\infty} 1 dx \text{ div}$$

$$\begin{aligned} \bullet \alpha \neq 1, \int_0^{\infty} |x|^{\alpha} dx &= \int_0^{\infty} e^{x \ln |x|} dx \\ &= \frac{1}{\ln |x|} e^{x \ln |x|} \Big|_0^{\infty} = \begin{cases} \infty & \textcircled{3} |x| > 1 \\ -\frac{1}{\ln |x|} & \textcircled{3} |x| < 1 \end{cases} \end{aligned}$$

Hence, I_{α} conv for $|x| < 1$ & div for $|x| > 1$.

Ex V. let $\alpha > 0$ and $n \in \mathbb{N}$.

$$1. \text{S.T. } \lim_{n \rightarrow \infty} \int_{2\pi n}^{2\pi n + \frac{\pi}{4}} x^{\alpha} \cos x dx = \infty$$

$$\star f(x) = \textcircled{1} x^{\alpha} \text{ cont on } [2\pi n, 2\pi n + \frac{\pi}{4}]$$

$$\star g(x) = \cos x \geq 0 \text{ \& cont (Hence integrable) on } [2\pi n, 2\pi n + \frac{\pi}{4}]$$

Then by 1st M.V.T $\exists 2\pi n \leq \xi_n \leq 2\pi n + \frac{\pi}{4}$ s.t.

$$\xi_n \xrightarrow{n \rightarrow \infty} \infty \text{ (Sand. Rule)}$$

$$\int_{2\pi n}^{2\pi n + \frac{\pi}{4}} x^{\alpha} \cos x dx = \xi_n^{\alpha} \int_{2\pi n}^{2\pi n + \frac{\pi}{4}} \cos x dx = e^{\frac{\alpha \ln \xi_n}{\sin \frac{\pi}{4}}} \xrightarrow{n \rightarrow \infty} \infty.$$

2. Deduce the nature of $\int_1^{\infty} x^{\alpha} \cos x dx$.

$$\int_{2\pi n}^{2\pi n + \frac{\pi}{4}} x^{\alpha} \cos x dx \xrightarrow{n \rightarrow \infty} \infty \Rightarrow \int_1^{\infty} x^{\alpha} \cos x dx \text{ div (Cauchy criterion)} \textcircled{5}$$

EXVI. let $I_n = \int_0^{\pi/2} \cos^{2n} t \, dt \quad \forall n \in \mathbb{N}$.

1. Show that $2n I_n = (2n-1) I_{n-1}$ for $n \geq 1$.

$$I_n = 2 \int_0^{\pi/2} \underbrace{\cos t}_{u'} \underbrace{\cos^{2n-1} t}_{v} dt \quad \left| \begin{array}{l} \textcircled{1} u = \sin t \text{ \& } v \in C([0, \pi/2]) \text{ (1 Bonus)} \\ \textcircled{1} v' = (2n-1) \cos^{2n-2} t (-\sin t) \end{array} \right.$$

$$\stackrel{\textcircled{1}}{=} 2 \left[\sin t \cos^{2n-1} t \Big|_0^{\pi/2} + (2n-1) \int_0^{\pi/2} \underbrace{\sin^2 t}_{1-\cos^2 t} \cos^{2n-2} t \, dt \right]$$

$$= (2n-1) \left[2 \int_0^{\pi/2} \cos^{2n-2} t \, dt - 2 \int_0^{\pi/2} \cos^{2n} t \, dt \right]$$

$$\stackrel{\textcircled{1}}{=} (2n-1) [I_{n-1} - I_n]$$

$\Downarrow$

$$2n I_n = (2n-1) I_{n-1}$$

2. Determine the value of I_n in terms of n .

$$I_n = \frac{2n-1}{2n} I_{n-1} \quad \text{for } n \geq 1$$

$\Downarrow \textcircled{3}$

$$I_n = \frac{2n-1}{2n} \cdot \frac{2n-3}{2n-2} \cdot \frac{2n-5}{2n-4} \cdots \frac{7}{8} \cdot \frac{5}{6} \cdot \frac{3}{4} \cdot \frac{1}{2} I_0$$

with $I_0 = 2 \int_0^{\pi/2} dt = 2 \cdot \pi/2 = \pi \cdot \textcircled{2}$

3. Show that $\int_0^1 \frac{x^{n-1/2}}{\sqrt{1-x}} dx = I_n$ (imp integ)

☆ $f(x) := \frac{x^{n-1/2}}{\sqrt{1-x}}; f \in C([0,1[) \subseteq R_{loc}([0,1[)$

☆ take $x = \cos^2 t = u(t)$ ②

$$\begin{aligned} \int_0^1 \frac{x^{n-1/2}}{\sqrt{1-x}} dx &\stackrel{\textcircled{1}}{=} \int_{\frac{\pi}{2}}^0 \frac{\cos^{2n-1} t}{\sqrt{1-\cos^2 t}} \cdot 2 \cos t (-\sin t) dt \stackrel{\textcircled{1}}{=} \\ &\stackrel{\textcircled{1}}{=} 2 \int_0^{\pi/2} \cos^{2n} t dt = I_n \end{aligned}$$

* $u \in C^1$ ($\overset{\textcircled{1 \text{ Bonus}}}{]0, \frac{\pi}{2}[}$) & invertible ($\overset{\textcircled{1 \text{ Bonus}}}{u' < 0}$ on $]0, \frac{\pi}{2}[$)

* f is cont on $u(\overset{\textcircled{1 \text{ Bonus}}}{]0, \frac{\pi}{2}[}) = [0, 1[$.